

The Geometry Garden — A Place to Discover

Name: _____ Date: _____ Class: _____

Build, observe, calculate, and explain.

Geometry World uses the familiar building and exploration of block-based 3D sandbox games to make geometric measurement visible. Count cubes in a layer, connect layers to multiplication, and compare strategies. Explaining and checking those calculations provides practice toward accurate, flexible calculation fluency.

An ungraded garden of eight discoveries: follow a level path from one unit cube to layers, equal volumes, composite shapes, nested cubes, and a hidden stepped monument. Use Activities for optional hints, travel, and a reflection journal. No quiz answers or scores are required. Explore at your own pace.

Lesson goals

1. Follow eight optional discoveries at your own pace
2. Measure real blocks and distinguish length, area, and volume
3. Compare shapes, layers, and materials using evidence
4. Walk around the screen wall to find the hidden stepped monument

Your lesson route

1. Garden Keeper · investigation
2. One cube, one unit · investigation
3. A row highlights length · investigation
4. A surface and a solid · investigation
5. Layers make volume · investigation
6. Three shapes, equal volume · investigation
7. Add the parts of an L · investigation
8. Look inside nested cubes · investigation
9. Discover the stepped monument · investigation

Print in portrait at 100% or Save as PDF. This companion follows the current lesson's tasks and question choices. Answer by hand; a drawing, words, or another clear representation can show your thinking. Use the world to gather observations before completing the paper independently.

Garden Keeper

Guide: **Garden Keeper** at world coordinates (0, 1, 3). Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Welcome to the Geometry Garden. No quizzes or scores are required. Follow the stone path, measure what interests you, and use Activities for eight optional discovery stops. Your journal is for observations and self-review. Near the far screen wall, the path turns around its end into the hidden garden. Take your time.

Self-review: Check the task requirements, label units, and explain how your evidence supports your result.

Plan / sketch · label dimensions and units

Measure / calculate · record the quantities and the operations you choose

Explain and review

How does your evidence support your result? Check it with a second method or describe a useful revision.

My mathematical reasoning and revisions

One cube, one unit

Guide: **Station 1 Guide** at world coordinates (4, 1, 0). Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Construction / investigation task

Find the single cyan cube. Predict its dimensions, then measure it. Explain what one cubic unit means.

This is a single unit cube — the building block of everything. Volume = 1 cubic unit. Press M to measure it!

Self-review: Check the task requirements, label units, and explain how your evidence supports your result.

Plan / sketch · label dimensions and units

Measure / calculate · record the quantities and the operations you choose

Explain and review

What makes this cube a useful unit for comparing all the larger models?

My mathematical reasoning and revisions

A row highlights length

Guide: **Station 2 Guide** at world coordinates (10, 1, 0). Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Construction / investigation task

Measure the five-cube cyan row. Compare it with the single cube, then imagine a second identical row beside it.

This row highlights length: five unit cubes sit side by side. It is still a three-dimensional block model, with length 5, width 1, and height 1. Measure it to find volume 5 cubic units. What would happen if you added another row?

Self-review: Check the task requirements, label units, and explain how your evidence supports your result.

Plan / sketch · label dimensions and units

Measure / calculate · record the quantities and the operations you choose

Explain and review

Which dimensions would change when you added a second row, and which would stay the same?

My mathematical reasoning and revisions

A surface and a solid

Guide: **Station 3 Guide** at world coordinates (18, 1, 0). Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Construction / investigation task

Study the top of the gold 5 by 3 model. Compare its 15-square-unit top surface with the volume of the one-layer solid.

The top surface of this model is a 5 by 3 rectangle, with area 15 square units. The model is also one cube thick, so its volume is 15 cubic units. The numbers match here, but square units describe a surface and cubic units describe the solid. Compare both ideas as you measure.

Self-review: Check the task requirements, label units, and explain how your evidence supports your result.

Plan / sketch · label dimensions and units

Measure / calculate · record the quantities and the operations you choose

Explain and review

When do the same numbers describe different measurements?

My mathematical reasoning and revisions

Layers make volume

Guide: **Station 4 Guide** at world coordinates (26, 1, 0). Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Construction / investigation task

Compare the three-layer cyan prism with the one-layer gold rectangle. Predict, then measure, how the volume changes.

This prism stacks three layers of 15 cubes. Its dimensions are 5 by 3 by 3, and its volume is 45 cubic units. Compare it with the one-layer gold model. What changes when you add layers while keeping the same base?

Self-review: Check the task requirements, label units, and explain how your evidence supports your result.

Plan / sketch · label dimensions and units

Measure / calculate · record the quantities and the operations you choose

Explain and review

How can you calculate the volume without counting all 45 cubes individually?

My mathematical reasoning and revisions

Three shapes, equal volume

Guide: **Convergence Guide** at world coordinates (36, 1, 4). Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Construction / investigation task

Find all three gold models near the crossing. Measure each separately and record a different set of dimensions for each.

Three gold shapes have volume 24: a 12 by 1 by 2 long slab, a 4 by 3 by 2 compact block, and a 2 by 2 by 6 tower. Measure each separately. Different dimensions can produce the same amount of occupied space. Which would you choose for a narrow site, and why?

Self-review: Check the task requirements, label units, and explain how your evidence supports your result.

Plan / sketch · label dimensions and units

Measure / calculate · record the quantities and the operations you choose

Explain and review

Which model suits a narrow space, and which suits a height limit? Explain your choice.

My mathematical reasoning and revisions

Add the parts of an L

Guide: **Composition Guide** at world coordinates (37, 1, -5). Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Construction / investigation task

Predict the volume of the cyan-and-gold L-shaped solid. Calculate each colored part, add them, then measure the whole.

The cyan part of this L-shaped model is 5 by 3 by 3: 45 cubic units. The gold part is 3 by 3 by 3: 27 cubic units. They touch without overlapping, so the whole has volume 72. Measure the connected structure and use the material breakdown to compare the two parts.

Self-review: Check the task requirements, label units, and explain how your evidence supports your result.

Plan / sketch · label dimensions and units

Measure / calculate · record the quantities and the operations you choose

Explain and review

Why would multiplying only the whole bounding dimensions overcount this L-shaped solid?

My mathematical reasoning and revisions

Look inside nested cubes

Guide: **Nested Cube Guide** at world coordinates (47, 1, -6). Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Construction / investigation task

Look through the glass shell at the gold cube. Calculate the outer cube, inner cube, and glass-only shell; then compare with Measure.

A 5 by 5 by 5 glass shell surrounds a gold 3 by 3 by 3 cube. The outside bounding cube contains 125 unit positions. Its inner cube uses 27, leaving 98 glass cubes in the shell. The gold now fills the cavity. Measure the touching glass-and-gold model as one structure: 125 total, with 98 glass and 27 gold in the material breakdown.

Self-review: Check the task requirements, label units, and explain how your evidence supports your result.

Plan / sketch · label dimensions and units

Measure / calculate · record the quantities and the operations you choose

Explain and review

What would the occupied volume become if the inner gold cube were removed? Explain without changing the protected model.

My mathematical reasoning and revisions

Discover the stepped monument

Guide: **Hidden Garden Sage** at world coordinates (53, 1, 10). Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Construction / investigation task

Follow the stone turn around the screen wall. Count the five layer areas of the hidden stepped monument and compare their sum with Measure.

You followed the path around the wall and found the stepped monument. Its five layers contain 45, 21, 15, 3, and 1 cubes, for a total of 85 cubic units. This stepped solid is an approximation, not an exact pyramid. Measure its occupied volume, then compare it with the larger bounding box. Why does the bounding box include so much empty space?

Self-review: Check the task requirements, label units, and explain how your evidence supports your result.

Plan / sketch · label dimensions and units

Measure / calculate · record the quantities and the operations you choose

Explain and review

What did this final discovery show you about measuring a shape with empty corners?

My mathematical reasoning and revisions

Connect the ideas

Choose one model or problem from this lesson. Explain how its representation connects to your calculations.

Model / diagram · show cubes, layers, dimensions, or a decomposition that fits the task

First calculation strategy · show the steps and units

Check with another strategy · explain why it agrees or what you revised

Reflection · what became easier and what do you still want to investigate?